

OmniVision Serial Camera Control Bus (SCCB) Functional Specification

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Revision Number	Date	Revision
1.0	06/07/00	Initial Release
1.01	06/08/00	Nomenclature change entire document - SIO1 changed to SIO_C, SIO0 changed to SIO_D, SCS_ changed to SCCB_E
2.0	03/08/02	Inclusion of Section 3.5 documenting the 2-wire master/slave implementation where SCCB_E is not available in the CAMERACHIP™ sensor
2.1	02/26/03	Incorporated into new template
2.2	06/25/07	In subsection 2.2, changed last sentence from "The minimum of t_{CYC} is 10 μ s" to "Typical for t_{CYC} is 10 μ s." In Table 5-1, added column for Typ values and moved t_{CYC} value of 10 μ s from Min value to Typ value.

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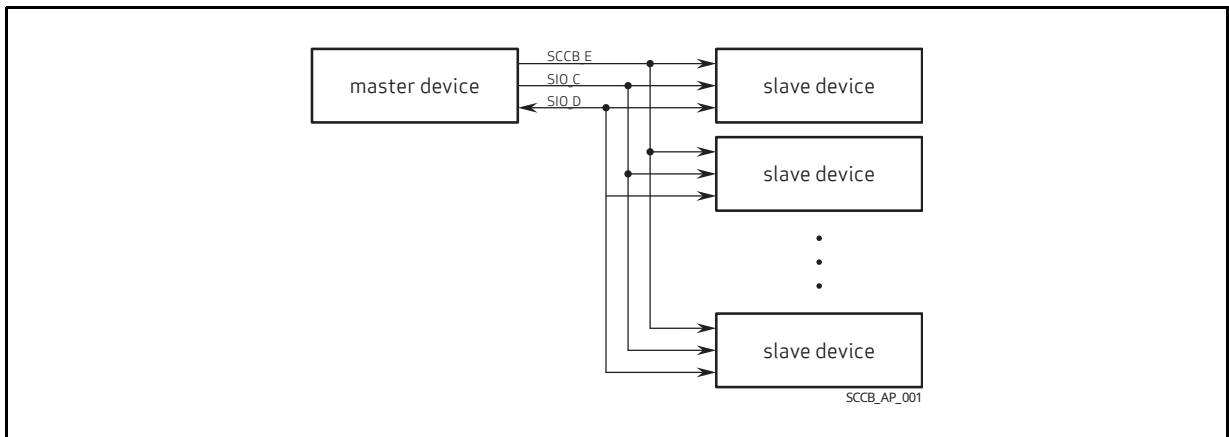
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1 Overview

OmniVision Technologies, Inc. has defined and deployed the Serial Camera Control Bus (SCCB), a 3-wire serial bus, for control of most of the functions in OmniVision's family of CAMERACHIP™ sensors. In reduced pin package parts, the SCCB operates in a modified 2-wire serial mode.

OmniVision CAMERACHIP sensors will only operate as slave devices and the companion back-end interface must assert as the master. One SCCB master device can be connected to the SCCB to control at least one SCCB slave device. An optional suspend-control signal provides the capability for the SCCB master device to power down the SCCB system. Refer to [Figure 1-1](#) for the SCCB functional diagram illustrating the 3-wire connection.

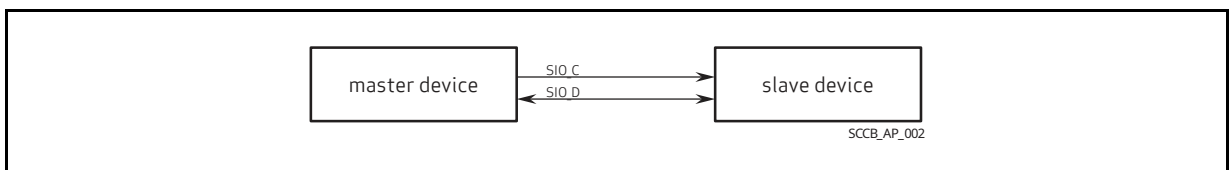
Figure 1-1 SCCB Functional Block Diagram



1.1 2-Wire SCCB Interface

The modified 2-wire implementation allows for a SCCB master device to interface with only one slave device. This 2-wire application is implemented in the CAMERACHIP sensors with reduced pin packages where the SCCB_E signal is not available externally. Refer to [Figure 1-2](#) for the functional diagram of the 2-wire implementation for the SCCB interface.

Figure 1-2 2-Wire SCCB Functional Block Diagram



The 2-wire implementation requires one of the following two master control methods in order to facilitate the SCCB communication.

1. In the first instance, the master device must be able to support and maintain the data line of the bus in a tri-state mode.
2. The alternate method if the master cannot maintain a tri-state condition of the data line is to drive the data line either high or low and to note the transition there to assert communications with the slave CAMERACHIP sensor.

2 Pin Functions

Refer to [Table 2-1](#) and [Table 2-2](#) for pin descriptions of the master and slave devices, respectively, used in SCCB communications.

Table 2-1. Master Device Pin Descriptions

Signal Name	Signal Type	Description
SCCB_E ^a	Output	Serial Chip Select Output - master drives SCCB_E at logical 1 when the bus is idle. <u>Drives at logical 0 when the master asserts transmissions or the system is in Suspend mode.</u>
SIO_C	Output	Serial I/O Signal 1 Output - <u>master drives SIO_C at logical 1 when the bus is idle.</u> Drives at logical 0 and 1 when SCCB_E is driven at 0. Drives at logical 0 when the system is Suspend mode.
SIO_D	I/O	Serial I/O Signal 0 Input and Output - remains floating when the bus is idle and drives to logical 0 when the system is in Suspend mode.
PWDN	Output	<u>Power down output</u>

a. Where SCCB_E is not present on the CAMERACHIP sensor, this signal is by default enabled and held high.

Table 2-2. Slave Device Pin Descriptions

Signal Name	Signal Type	Description
SCCB_E ^a	Input	Serial Chip Select Input - input pad can be shut down when the system is in Suspend mode.
SIO_C	Input	Serial I/O Signal 1 Input - input pad can be shut down when the system is in Suspend mode.
SIO_D	I/O	Serial I/O Signal 0 Input and Output - input pad can be shut down when the system is in Suspend mode.
PWDN	Input	Power down input

a. Where SCCB_E is not present on the CAMERACHIP sensor, this signal is by default enabled and held high.

2.1 SCCB_E Signal

The SCCB_E signal is a single-directional, active-low, control signal that must be driven by the master device. It indicates the start or stop of the data transmission. A high-to-low transition of the SCCB_E indicates a start of a transmission, while the low-to-high transition of the SCCB_E indicates a stop of a transmission. SCCB_E must remain at logical 0 during a data transmission. A logical 1 of SCCB_E indicates that the bus is idle.

2.2 SIO_C

The SIO_C signal is a single-directional, active-high, control signal that must be driven by the master device. It indicates each transmitted bit. The master must drive SIO_C at logical 1 when the bus is idle. A data transmission starts when SIO_C is driven at logical 0 after the start of transmission. A logical 1 of SIO_C during a data transmission indicates a single transmitted bit. Thus, SIO_D can occur only when SIO_C is driven at 0. The period of a single transmitted bit is defined as t_{CYC} as shown in [Figure 3-8](#). Typical for t_{CYC} is 10 μ s.

2.3 SIO_D

The SIO_D signal is a bi-directional data signal that can be driven by either master or slave devices. It remains floating, or tri-state, when the bus is idle. Maintenance of the signal is the responsibility of both the master and slave devices in order to avoid propagating an unknown bus state.

Bus float and contention are allowed during transmissions of Don't-Care or NA bits. The definition of the Don't-Care bit is described in [Section 3.2.3](#). The master must avoid propagating an unknown bus state condition when the bus is floating or conflicting. A conflict-protection resistor is required to reduce static current when the bus conflicts. The connection of the conflict-protection resistor is shown in [Figure 4-2](#).

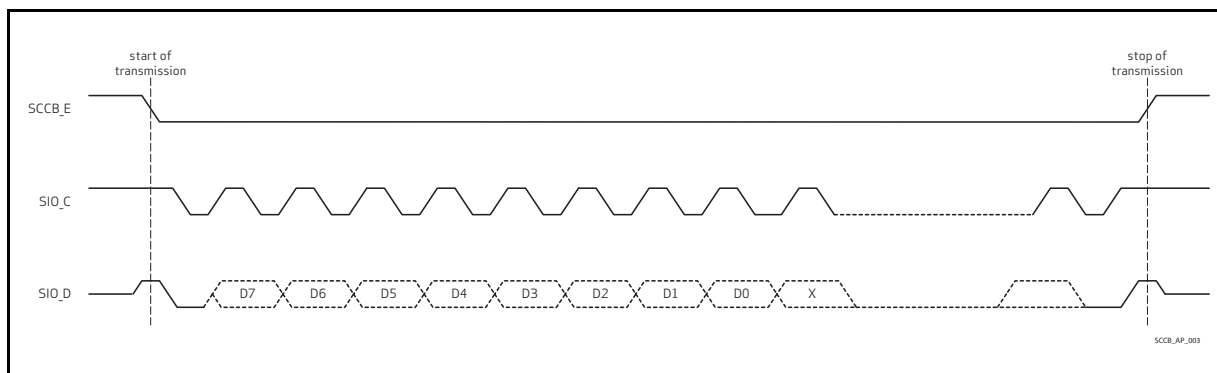
A single-bit transmission is indicated by a logical 1 of SIO_C. SIO_D can occur only when SIO_C is driven at logical 0. However, an exception is allowed at the beginning and the end of a transmission. During the period that SCCB_E is asserted and before SIO_C goes to 0, SIO_D can be driven at 0. During the period that SIO_C goes to 1 and before SCCB_E is de-asserted, SIO_D can also be driven at 0.

3 Data Transmission

3.1 3-Wire Data Transmission

A graphic overview of the SCCB 3-wire data transmission is shown in [Figure 3-1](#). The SCCB protocol allows for bus float and contention during data transmissions. Writing data to slaves is defined as a write transmission, while reading data from slaves is defined as a read transmission.

Figure 3-1 3-Wire Data Transmission Timing Diagram

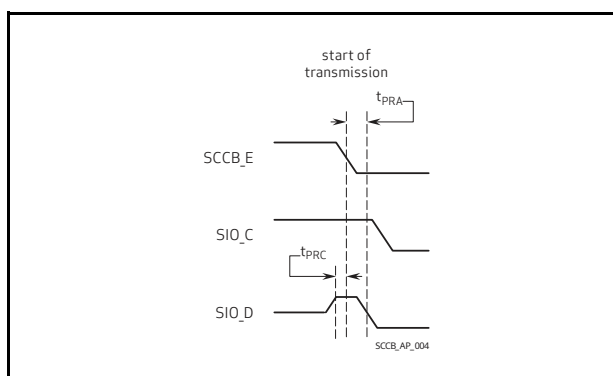


3.1.1 Start of Data Transmission

The start of data transmission in the 3-wire implementation is indicated by a high-to-low transition of SCCB_E. Before asserting SCCB_E, the master must drive SIO_D at logical 1. This will avoid propagating an unknown bus state before the transmission of data. After de-asserting SCCB_E, the master must drive SIO_D at 1 for a defined period again to avoid unknown bus state propagation. This period, t_{PSA} , is defined as the post-active time of SCCB_E and has a minimum value of 0 μs .

Two timing parameters are defined for the start of transmission, t_{PRC} and t_{PRA} . The t_{PRC} is defined as the pre-charge time of SIO_D. This indicates the period that SIO_D must be driven at logical 1 prior to assertion of SCCB_E. The minimum value of t_{PRC} is 15 ns. The t_{PRA} is defined as the pre-active time of SCCB_E. This indicates the period that SCCB_E must be asserted before SIO_D is driven at logical 0. The minimum value of t_{PRA} is 1.25 μs . The 3-wire start of transmission is shown in [Figure 3-2](#).

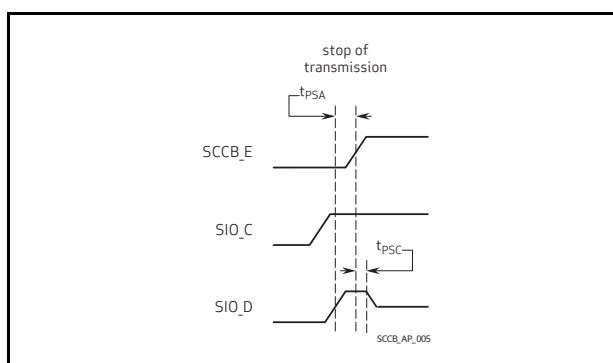
Figure 3-2 3-Wire Start of Data Transmission



3.1.2 Stop of Data Transmission

A stop of data transmission is indicated by a low-to-high transition of SCCB_E. Two timing parameters are defined for the stop of transmission, t_{PSC} and t_{PSA} . The t_{PSC} is defined as post-charge time of SIO_D. It indicates the period that SIO_D must remain at logical 1 after SCCB_E is de-asserted. The minimum value of t_{PSC} is 15 ns. The t_{PSA} is defined as the post-active time of SCCB_E. It indicates the period that SCCB_E must remain at logical 0 after SIO_D is de-asserted. The minimum value of t_{PSA} is 0 ns. The 3-wire stop of transmission is shown in Figure 3-3.

Figure 3-3 3-Wire Stop of Data Transmission



3.2 Transmission Cycles

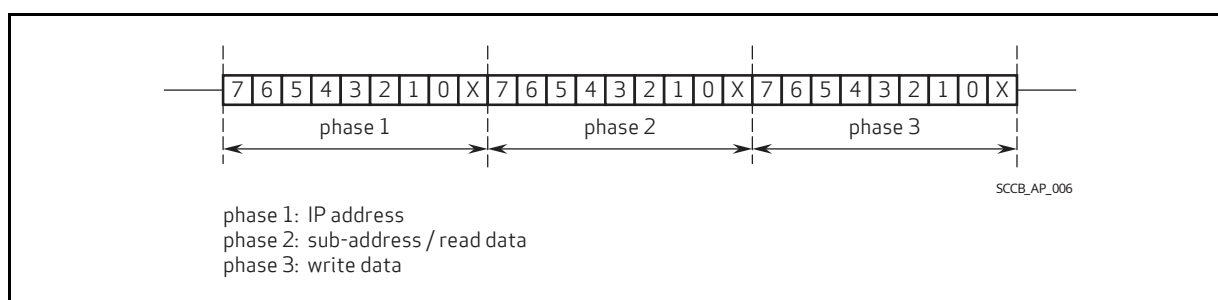
A basic element of a data transmission is called a phase. This section describes the three kinds of transmissions:

- [3-Phase Write Transmission Cycle](#)
- [2-Phase Write Transmission Cycle](#)
- [2-Phase Read Transmission Cycle](#)

3.2.1 Transmission Phases

A phase contains a total of 9 bits. The 9 bits consist of an 8-bit sequential data transmission followed by a ninth bit (see Figure 3-4). The ninth bit is a Don't-Care bit or an NA bit, depending on whether the data transmission is a write or read. The maximum number of phases that can be included in a transmission is three. The Most Significant Bit (MSB) is always asserted first for each phase.

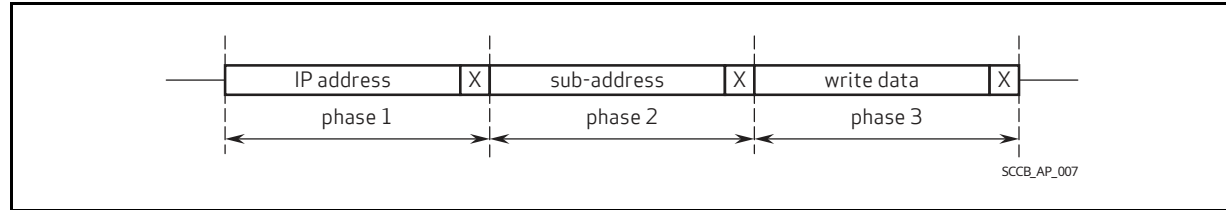
Figure 3-4 Transmission Phases



3.2.1.1 3-Phase Write Transmission Cycle

The 3-phase write transmission cycle (see Figure 3-5) is a full write cycle such that the master can write one byte of data to a specific slave(s). The ID address identifies the specific slave that the master intends to access. The sub-address identifies the register location of the specified slave. The write data contains 8-bit data that the master intends to overwrite the content of this specific address. The ninth bit of the three phases will be Don't-Care bits.

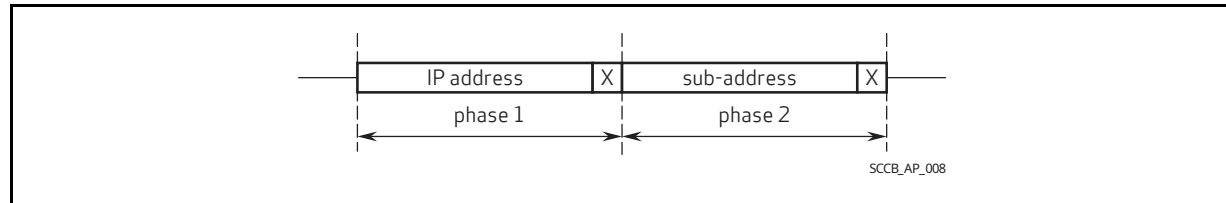
Figure 3-5 3-Phase Write Transmission Cycle



3.2.1.2 2-Phase Write Transmission Cycle

The 2-phase write transmission cycle is followed by a 2-phase read transmission cycle. The purpose of issuing a 2-phase write transmission cycle (see Figure 3-6) is to identify the sub-address of some specific slave from which the master intends to read data for the following 2-phase read transmission cycle. The ninth bit of the 2-phase write transmission will be Don't-Care bits.

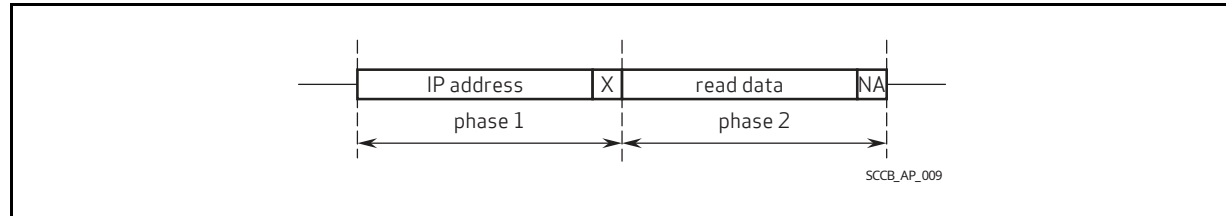
Figure 3-6 2-Phase Write Transmission Cycle



3.2.1.3 2-Phase Read Transmission Cycle

There must be either a 3-phase or a 2-phase write transmission cycle asserted ahead of a 2-phase read transmission cycle. The 2-phase read transmission cycle (see Figure 3-7) has no ability to identify the sub-address. The 2-phase write transmission cycle contains read data of 8 bits and a ninth Don't-Care bit or NA bit. The master must drive the NA bit at logical 1.

Figure 3-7 2-Phase Read Transmission Cycle



3.2.2 Phase Descriptions

The following sections describe the individual phases found in the various transmission cycles.

3.2.2.1 Phase 1 — ID Address

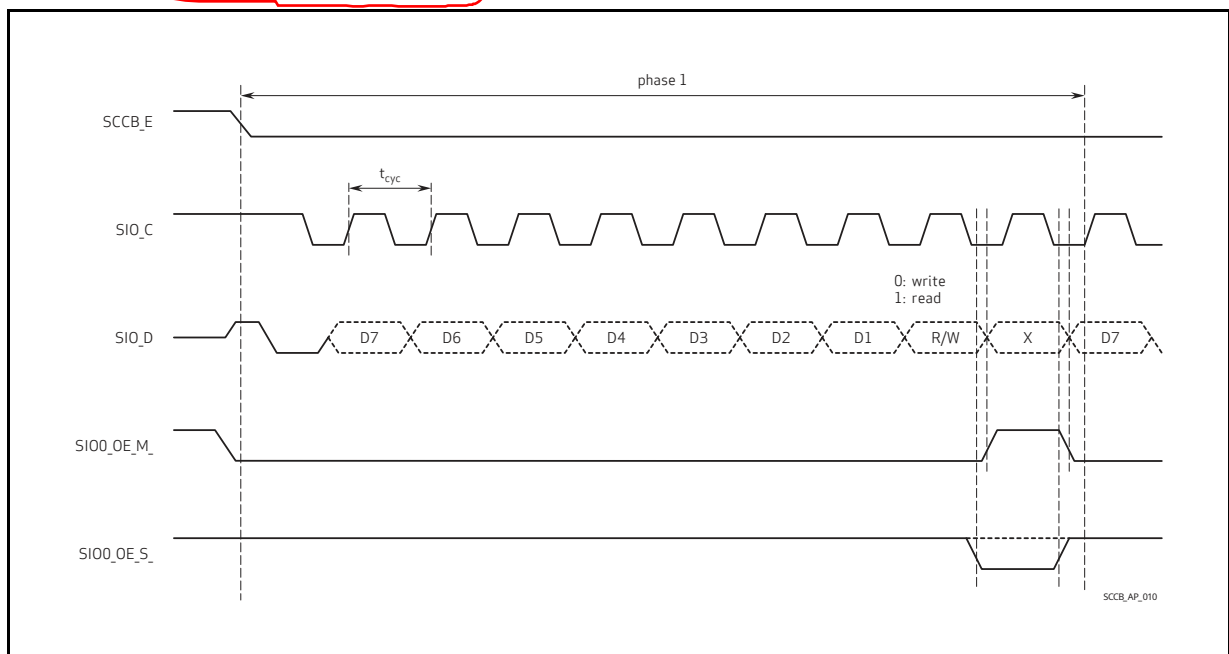
Phase 1 is asserted by the master to identify the selected slave to which the data is read or written. Each slave has a unique ID address. The ID address is comprised of seven bits, ordered from bit 7 to bit 1, and can identify up to 128 slaves. The eighth bit, bit 0, is the read/write selector bit that specifies the transmission direction of the current cycle. A logical 0 represents a write cycle and a logical 1 represents a read cycle.

The ninth bit of Phase 1 must be a Don't-Care bit. SIO_D_OE_M_ and SIO_D_OE_S_ shown in [Figure 3-8](#) are internal active-low, I/O-enabled signals in the master and slave(s), respectively. SIO_D_OE_S_ transaction occurs before the transition of SIO_D_OE_M_, as shown in [Figure 3-8](#). The master asserts the ID address, but de-asserts the ninth bit (Don't-Care bit). The master must mask the input of SIO_D during the period of the Don't-Care bit and force the input to 0 to avoid propagating an unknown bus state. The master continues asserting the following phases regardless of the response to the Don't-Care bit by the slave(s).

The SIO_OE_S is controlled by the slave(s) and may remain at logical 1 or be driven at logical 0. The bus may be in a floating or conflicting status during the transmission of the Don't-Care bit. In this case, it is the slave's responsibility to avoid propagating an unknown bus state.

A detailed description of the Don't-Care bit is described in [Section 3.2.3](#).

Figure 3-8 **Phase 1 — ID Address**

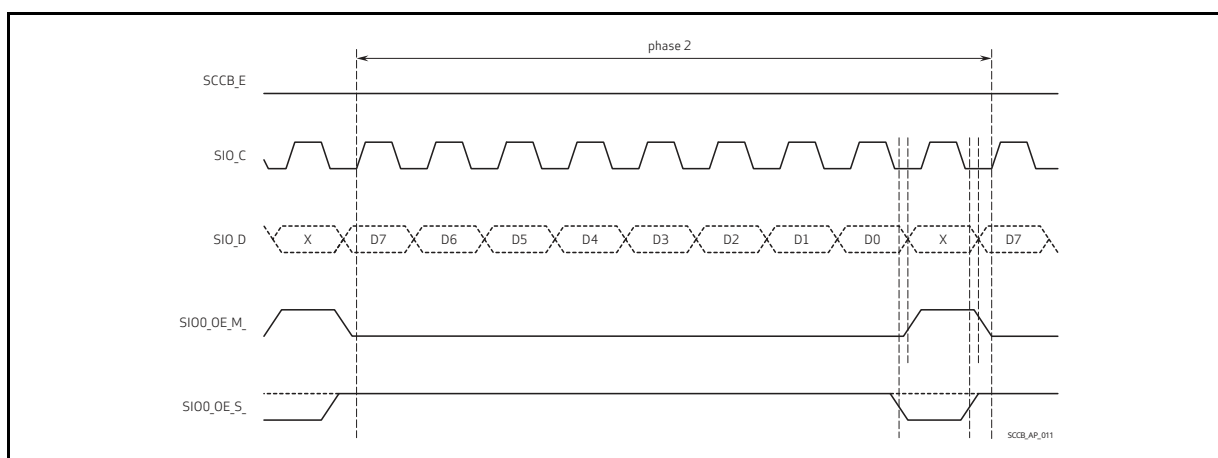


3.2.2.2 Phase 2 — Sub-address/Read Data

Either the master or the slave(s) may assert a phase 2 transmission. A phase 2 transmission asserted by the master identifies the sub-address of the slave(s) the master intends to access. A phase 2 transmission asserted by the slave(s) indicates the read data that the master will receive. The slave(s) recognize the sub-address of this read data according to the previous 3-phase or 2-phase write transmission cycles.

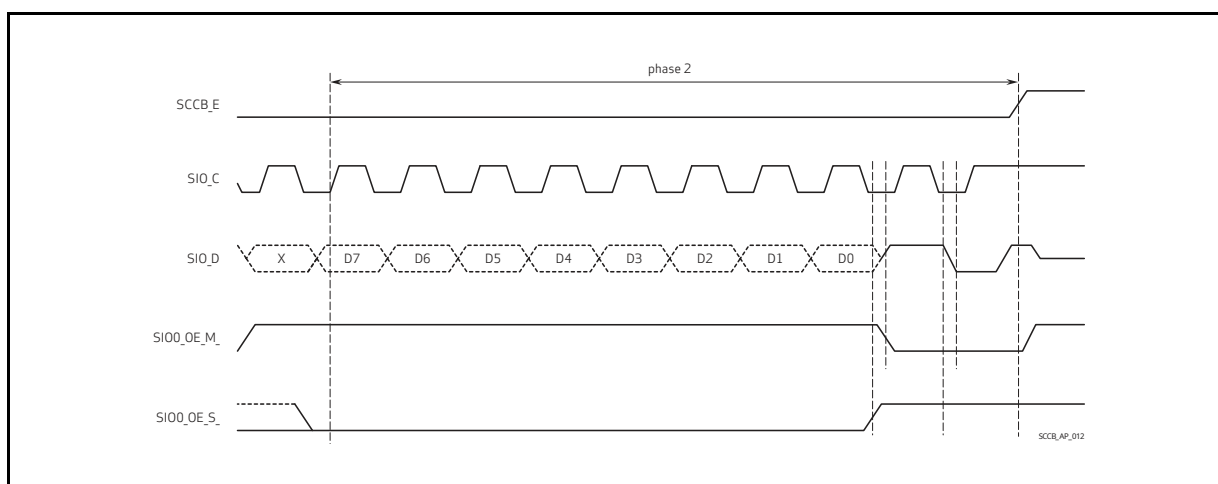
The ninth bit is defined as a Don't-Care bit when the master asserts phase 2. SIO_D_OE_M_ and SIO_D_OE_S_ are the same as those defined under [Section 3.2.2.1](#). The detailed timing is illustrated in [Figure 3-9](#).

Figure 3-9 Phase 2 — Sub-address (3-Phase Write Transmission)



The ninth bit is defined as an NA bit when the slave(s) assert the phase 2 transmission. SIO_D_OE_M_ is de-asserted from the ninth bit of phase 1 and re-asserted for the NA bit. The master is responsible for driving SIO_D at logical 1 during the period of the NA bit. Concurrently, SIO_D_OE_S_ is asserted. The selected slave is responsible for driving SIO_D during the read data period. Since SIO_D_OE_S_ is de-asserted before SIO_D_OE_M_ is asserted during the period of the NA bit, bus float of SIO_D occurs when the master tries to drive the NA bit. The detailed timing is shown in [Figure 3-10](#).

Figure 3-10 Phase 2 — Read Data (2-Phase Read Transmission)

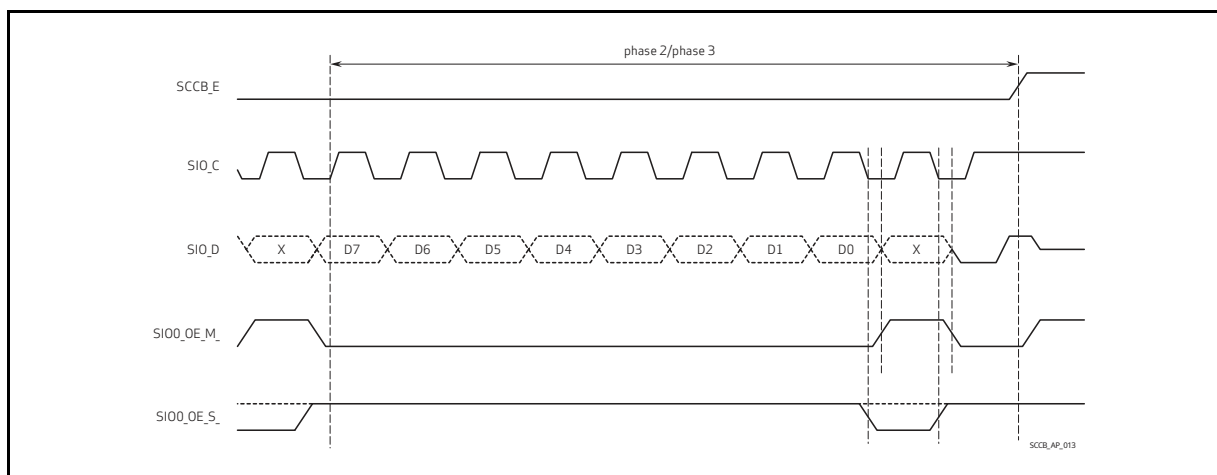


3.2.2.3 Phase 3 — Write Data

Only the master may assert the phase 3 transmission. The phase 3 transmission contains the actual data the master intends to write to the slave(s). The timing diagram shown in [Figure 3-11](#) applies to both Phase 2 sub-address write transmissions and Phase 3 write data transmissions.

The ninth bit of the phase 3 transmission is defined as a Don't-Care bit since the master is asserting the transmission. SIO_D_OE_M_ and SIO_D_OE_S_ are the same as those defined for a phase 1 transmission.

Figure 3-11 Phase 2 Sub-address Write Transmission/Phase 3 Write Data Transmission



3.2.3 Don't-Care Bit

The Don't-Care bit is the ninth bit of a master-issued transmission (ID address, sub-address, and write data). The master will continue to assert transmission phases until the transmission cycle is complete. The master also assumes that there is no transmission error during data transmissions. The purpose of the Don't-Care bit is to indicate the completion of the transmission.

When there is more than one slave on the bus, the slave(s) may respond to the Don't-Care bit in one of two ways. If slave 1 is selected and data is written to this specific slave, slave 1 will drive SIO_D to logical 0 for the Don't-Care bit. In this case, the SIO_D signal may conflict at the beginning of the Don't-Care bit, while it may be floating at the end of the Don't-Care bit (see [Figure 3-12](#)).

Alternately, it is possible that the slave(s) do not respond to the Don't-Care bit of the current phase. In this situation, the SIO_D bus remains at float for the whole Don't-Care bit.

The master does not check for transmission errors during data transmissions. There is a provision for the slave(s) to record the status of the Don't-Care bit in an internal register as shown in the following example:

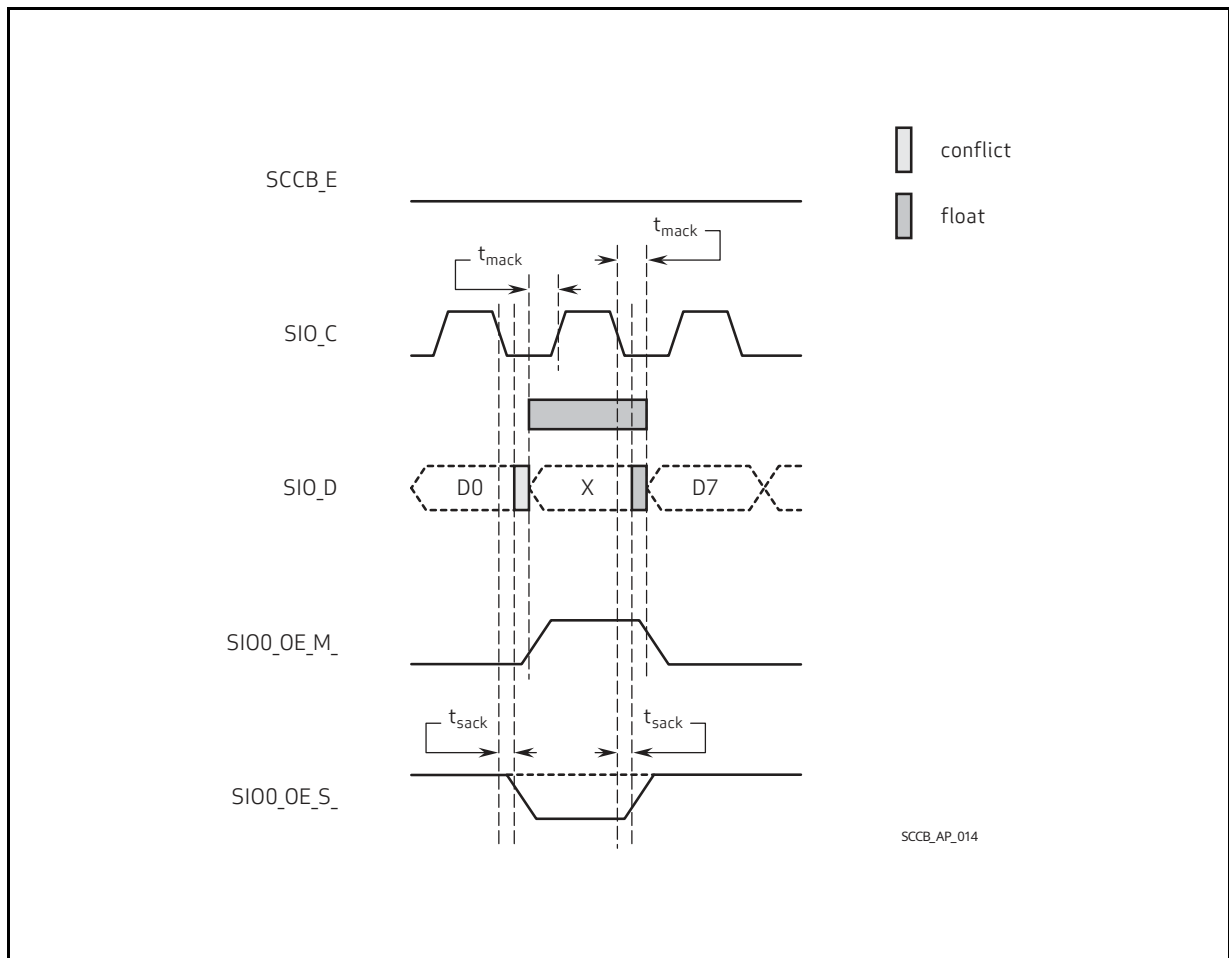
A slave(s) has defined a 1 byte register as the Don't-Care Status register. The default value of the Don't-Care Status register is defined as 55. Assuming there are no errors during the data transmission, this register value will remain unchanged. If the slave does not receive the Don't-Care bit, the register value will change to 54.

The master may query the Don't-Care Status register to determine if there has been a transmission of data. The master will issue an additional read transmission to the Don't-Care Status register in the target slave to check the value and, subsequently, determine if an error has occurred. This scheme will not determine an error if the entire SCCB circuit has been corrupted.

SIO_D_OE_M_ can be de-asserted and re-asserted during the Don't-Care bit transmission only when SIO_C is driven to logical 0. The t_{mack} is defined as the period of de-assertion of SIO_D_OE_M_ prior to the low-to-high transition of SIO_C during the Don't-Care bit transmission. The period of re-assertion of SIO_D_OE_M_ after the high-to-low transition of SIO_C is also defined as t_{mack} . The minimum value of t_{mack} is 1.25 μ s.

If a slave intends to respond to the Don't-Care bit, SIO_D_OE_S_ can be asserted and de-asserted during the Don't-Care bit transmission only when SIO_C is driven to logical 0. The t_{sack} is defined as the period of assertion of SIO_D_OE_S_ occurring after the high-to-low transition of SIO_C at the beginning of the Don't-Care bit transmission. The period of de-assertion of SIO_D_OE_S_ occurring after the high-to-low transition at the end of the Don't-Care transmission is also defined as t_{sack} . The minimum value of t_{sack} is 370 ns.

Figure 3-12 Don't-Care Bit

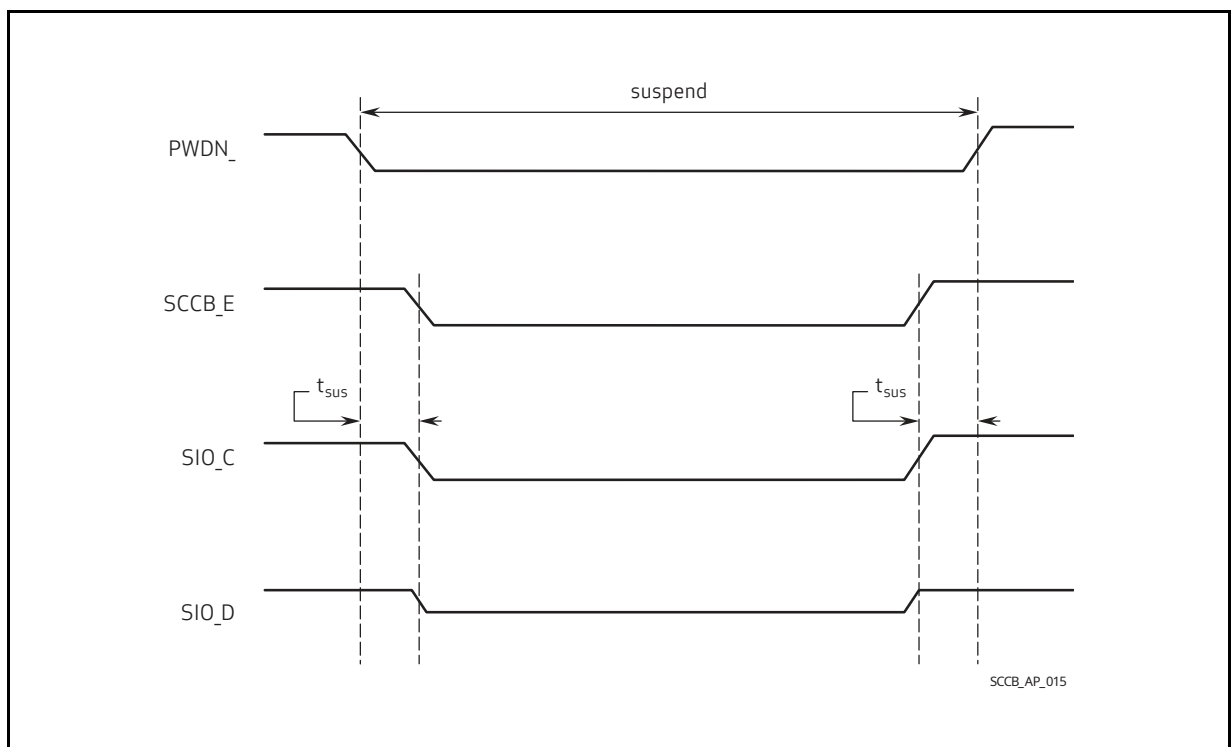


3.3 Suspend Mode

Suspend mode (see [Figure 3-13](#)) is determined by the dedicated PWDN_ pin of the master. This is achieved by the active-low output signal that specifies the suspension period as the master attempts to power down the system. During the suspension period, SCCB_E, SIO_C, and SIO_D are all driven to logical 0 by the master in order to avoid current leakage. There must be some time for PWDN_ to be asserted prior to and be de-asserted after the assertion of SCCB_E, SIO_D, and SIO_C. This parameter is defined as t_{sup} . The minimum value of t_{sup} is 50 ns. This scheme can prevent logical errors from occurring in SCCB slaves.

The PWDN pin in slaves has the opposite polarity of the PWDN_ pin of the master. Two control schemes for suspending the slave(s) are described in [Section 4.4](#).

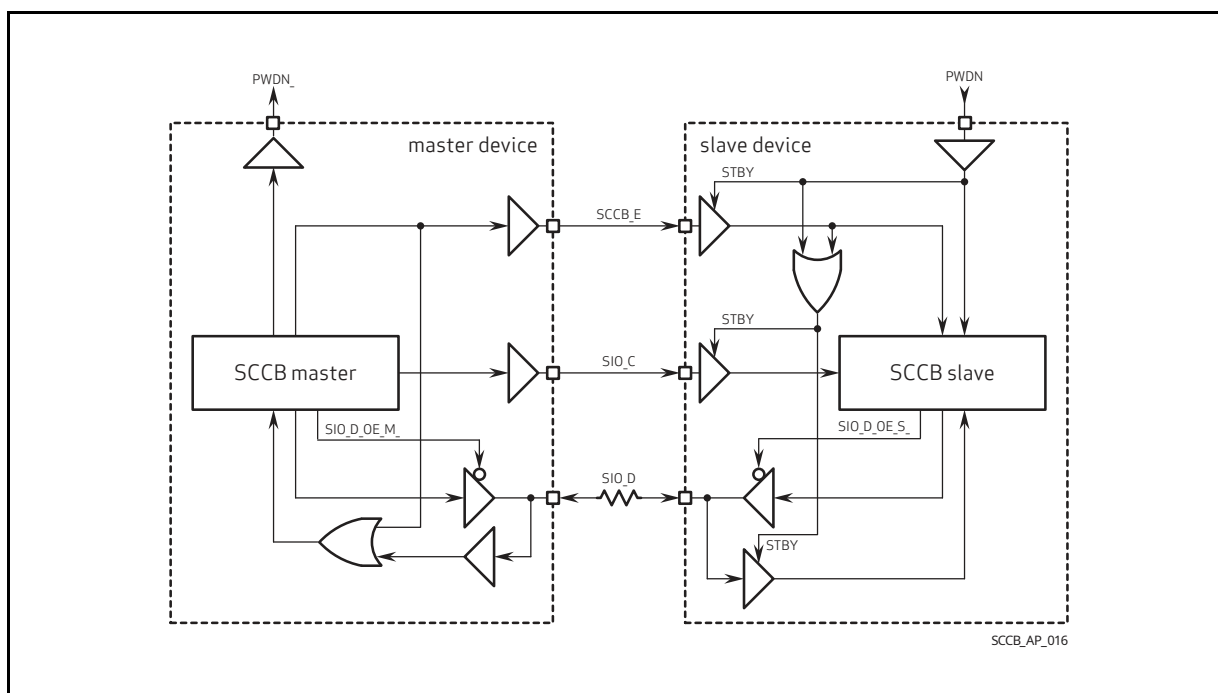
Figure 3-13 Suspend Mode



4 SCCB Structure

The structure of the SCCB system is shown in [Figure 4-1](#). This diagram illustrates the connection of one master with one slave. Multiple slaves may be connected on the same bus. A conflict-protection resistor of SIO_D is required for each slave. Connection of conflict-protection resistors for multiple slaves is illustrated in [Figure 4-3](#).

Figure 4-1 Block Diagram of the Master and Slaves



4.1 Master Device

The master device drives both SCCB_E and SIO_C signals, while either the master or slave(s) can drive the SIO_D signal. During the de-assertion of SCCB_E, the master must block the SIO_D input to avoid propagating unknown bus conditions due to bus float. During the Don't-Care bit transmission, the master must ignore the status of SIO_D and keep asserting the subsequent phases.

The PWDN_ is driven by the master to indicate the suspend mode cycle. As noted in [Section 4.4](#), there are two different ways to implement suspension circuits within the system.

4.2 Slave Devices

The slave(s) receive the SCCB_E and SIO_C signals from the master, while either the master or the slave(s) can drive SIO_D. Input pads of the SCCB_E, SIO_C, and SIO_D signals contain the standby (STBY) control terminal for reducing leakage current when the inputs are floating. Output terminals of those input pads are driven at logical 1 when STBY is asserted. This can avoid logical errors during suspend cycles.

PWDN controls STBY of SCCB_E. This means the output terminal of the SCCB_E input pad is driven at logical 1 during suspend mode cycles even though the master drives the input of SCCB_E at 0.

The STBY control terminals of both SIO_C and SIO_D are controlled by PWDN and SCCB_E. During suspend mode cycles and the de-assertion of SCCB_E, the output terminals of SIO_C and SIO_D input pads are both driven at logical 1. During the Don't-Care bit transmission, the slave(s) must avoid propagating unknown bus conditions.

4.3 Conflict-Protection Resistors

Incorporating series resistors between SIO_D output of the master and the SIO_D input of the slave(s) can avoid short circuits when bus contention occurs.

Figure 4-2 Conflict-Protection Resistor Connections

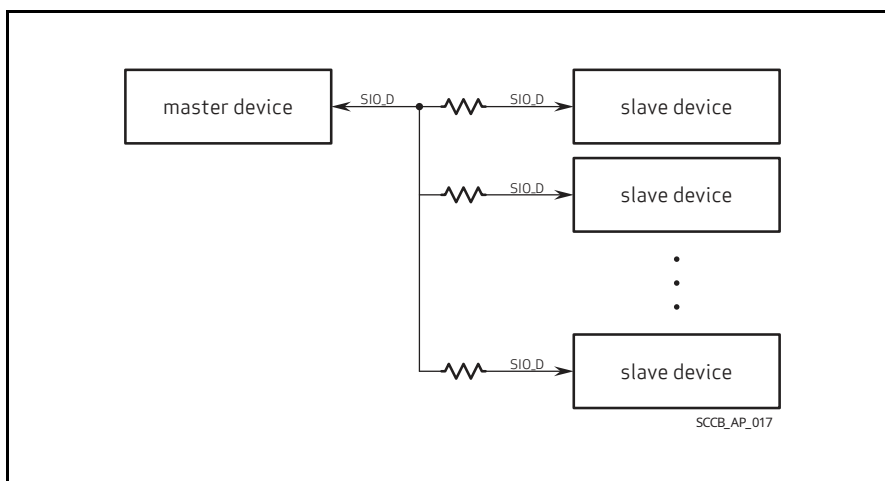
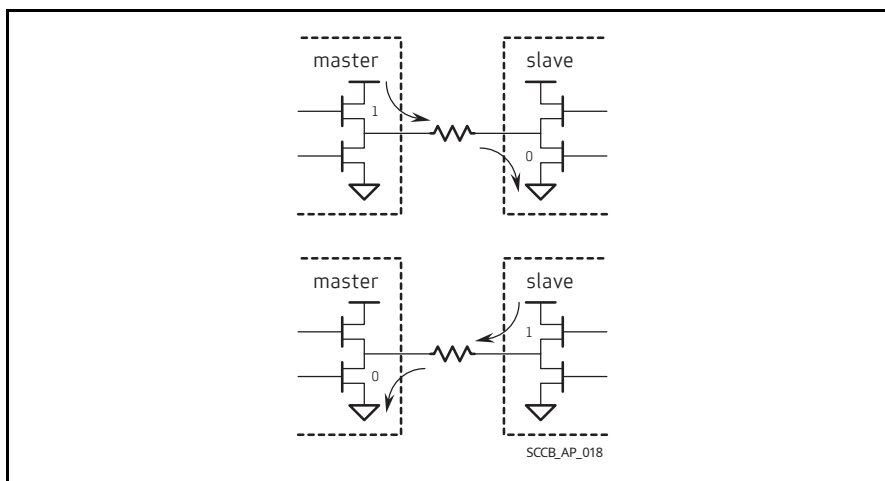


Figure 4-3 Conflict-Protection Resistors



4.4 Suspend Circuits

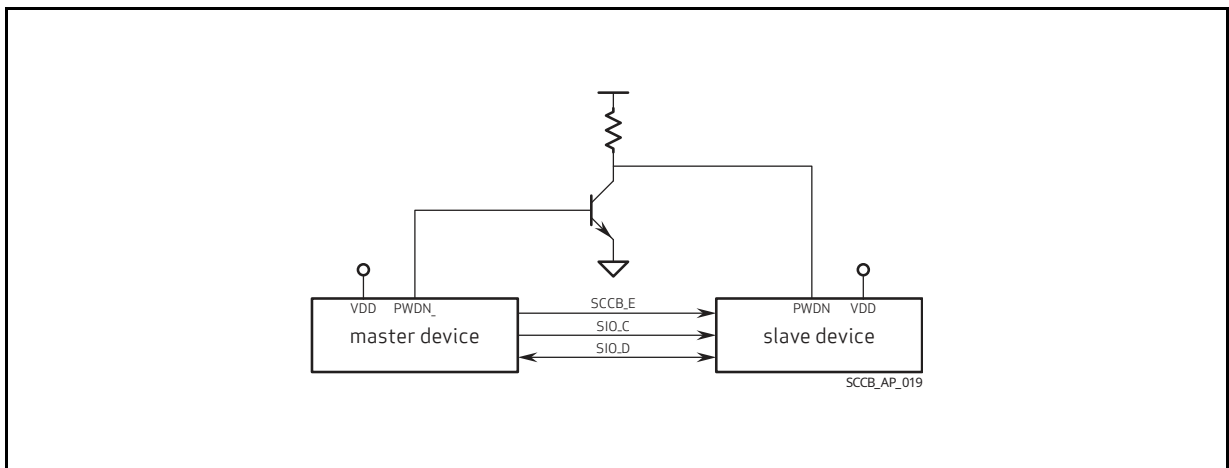
There are two methods of issuance of a bus suspend cycle:

- PWDN Mode
- Switch Mode

4.4.1 PWDN Mode

The power pads of the slave(s) are always connected to VDD. The PWDN_ signal from the master device needs to be inverted prior to connection to the slave(s) and the slave(s) circuit has an opposite polarity. During normal operations, PWDN_ of the master is driven at logical 1 and the NPN transistor is ON. In normal operation, the PWDN of the slave(s) is driven at logical 0. During the suspend mode cycle, PWDN_ is driven at 0 and the NPN transistor is OFF. During the suspend mode operation, the PWDN of the slave(s) is driven at 1. There is no leakage current during the suspend cycle.

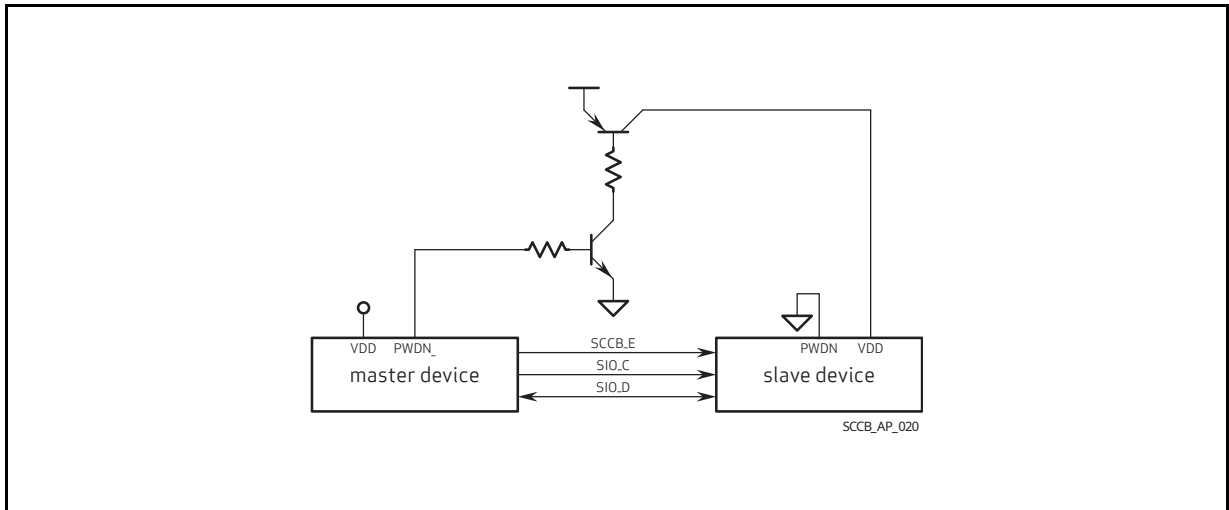
Figure 4-4 Suspend Circuit - PWDN Mode



4.4.2 Switch Mode

The PWDN circuit of the slave(s) is always connected to logical 0. A power switch circuit is required for each slave. The power of each slave is OFF during suspend mode cycles. In suspend mode operation, there is no leakage current present as no power is provided to the slave(s).

Figure 4-5 Suspend Circuit - Switch Mode



5 Electrical Characteristics

Table 5-1. SCCB Electrical Characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit
t_{cyc}	Single bit transmission cycle time			10		μs
t_{prc}	Pre-charge time of SIO_D		15			ns
t_{pra}	Pre-active time of SCCB_E		1.25			μs
t_{psc}	Post-charge time of SIO_D		15			μs
t_{psa}	Post-active time of SCCB_E		0			μs
t_{mack}	SIO_D_OE_M_ transition time		1.25			μs
t_{sack}	SIO_D_OE_S_ transition time		370			ns
t_{sup}	PWDN_ pre/post-charge time		50			ns

6 Terminology

<i>Don't-Care Bit:</i>	Ninth bit of a Write phase.
<i>ID Address:</i>	Unique address of each device on the bus. The master asserts the slave ID address to identify transmissions destined for the slave device(s).
<i>NA Bit:</i>	Ninth bit of a Read phase.
<i>Phases:</i>	A phase contains a total of 9 bits consisting of a sequential transmission of 8 data bits followed by a ninth Don't-Care or NA bit, depending on writes or reads.
<i>Read Phases:</i>	Phases that read data from slave(s).
<i>Read Transmissions:</i>	Master-asserted transmissions which read data from slave device(s).
<i>SCCB Data Transmissions:</i>	Transmissions consist of phases. All transmissions are initiated by the master device. Start and stop of a transmission in the 3-wire system are indicated by the signaling of SCCB_E. Start and stop of a transmission in the 2-wire system are indicated by signaling of the SIO_D.
<i>SCCB_E:</i>	Serial bus enable/disable signal, previously denoted as SCS_, SCCBB, and IICB in older documentation.
<i>SCCB Master Device:</i>	An SCCB device that can assert SCCB transmissions. Only one master is allowed in the system.
<i>SCCB Slave Device(s):</i>	SCCB device(s) that can respond to an asserted SCCB transmission. At least one slave can be connected to the system.
<i>SCCB System:</i>	System consists of one master and at least one slave.
<i>Serial Camera Control Bus: (SCCB)</i>	Typically, a 3-wire serial bus with an optional suspend-control signal. May be implemented in a 2-wire mode where required.
<i>SIO_C:</i>	Serial bus clock signal, previously denoted as SIO1 and SCL in older documentation.
<i>SIO_D:</i>	Serial bus data signal, previously denoted as SIO0 and SDA in older documentation.

<i>Sub-address:</i>	The master asserts the sub-address to indicate the specific slave function/location to be accessed.
<i>Suspend Mode:</i>	Master-asserted suspend periods of device and/or system suspension.
<i>Transmission Cycles:</i>	Transmission cycles include 3-phase write transmission cycle, 2-phase write transmission cycle, and 2-phase read transmission cycle.
<i>Write Transmissions:</i>	Master-asserted transmissions which write data to slave device(s).

Note:

- *All information shown herein is current as of the revision and publication date. Please refer to the OmniVision web site (<http://www.ovt.com>) to obtain the current versions of all documentation.*
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